

**Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of  
Selective Interference**

Jordan B Gunn

Department of Psychology, University of Cambridge

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### **Abstract**

Visuospatial tasks performed after trauma-film encoding, or after a later reminder of the film, can reduce intrusive memories while leaving voluntary memory relatively intact. This selective interference effect is often treated as evidence that intrusive and voluntary remembering depend on differently vulnerable memory representations. Here we simulate an alternative retrieved-context account using an emotional Context Maintenance and Retrieval (eCMR) model, asking whether context reinstatement, competitor learning, and retrieval control can generate the selective-interference pattern without separately vulnerable representations. In the model, film context remains active after encoding and can later be reinstated by reminders, so an interfering task performed in either state binds task representations to overlapping context. When later retrieval contexts cue both film representations and task competitors, uncontrolled retrieval from context cues is less likely to sample trauma-film items in intrusion-like retrieval, even though film-item strength is not reduced. Deliberate recall is less vulnerable when task goals and control processes reinstate film context and prioritize film items over competitors for retrieval; film-specific cues can also steer retrieval toward film context across test formats. Together, the simulations reproduce selective impairment of intrusion-like retrieval and specify when the intrusion-voluntary-memory dissociation should be more or less pronounced, helping to interpret heterogeneity across empirical paradigms. Selective interference is therefore compatible with a single retrieved-context system and is not by itself diagnostic of separate representational systems or selective trace weakening.

*Keywords:* intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

## **Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of Selective Interference**

### **Introduction**

Traumatic experience foregrounds a tension between durability and selectivity in episodic memory. Highly consequential events should be retained well enough in memory that critical details remain accessible long into the future. At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current cues, goals, and safety. Intrusive memories after trauma suggest that durable encoding can outpace regulatory control. The event returns without being deliberately sought and without being well matched to the demands of the present ([Brewin et al., 2010](#); [Ehlers & Clark, 2000](#); [Iyadurai et al., 2019](#); [Krans et al., 2009](#)). Understanding why such memories arise, and how they differ from voluntary memory for the same event, is therefore a shared target for both clinical science and basic memory theory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event ([Holmes & Bourne, 2008](#); [James et al., 2016](#)). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain available to deliberate retrieval while becoming more or less likely to return involuntarily. In such research, voluntary memory can be relatively preserved, and its relationship to intrusion frequency can be weak, such that the two measures dissociate ([Holmes et al., 2004](#); [James et al., 2016](#)). The same encoded episode can therefore support both uncontrolled and controlled access, and these modes can come apart empirically ([Holmes et al., 2004](#); [Iyadurai et al., 2019](#)).

Visuospatial interference tasks make this dissociation experimentally sharper. When brief tasks such as Tetris are performed after film encoding or after a later film reminder, subsequent intrusions can be reduced while voluntary memory for the same material shows little measurable impairment ([Asselbergs et al., 2023](#); [Deepprose et al., 2012](#); [Holmes et al., 2009, 2010](#); [James et al., 2015](#); [Kessler et al., 2020](#); [Lau-Zhu et al., 2019](#)). Even for memories of real-life traumatic

events, including motor vehicle accidents, emergency caesarean section, and war-related trauma, the same post-reminder intervention can reduce intrusions while leaving deliberate recall relatively intact ([Horsch et al., 2017](#); [Iyadurai et al., 2018](#); [Kanstrup et al., 2021](#)). At the same time, the evidence base is heterogeneous, and recent reviews and replications caution against treating the effect as settled or uniform ([Asselbergs et al., 2023](#); [Varma et al., 2024](#); [Wessel et al., 2025](#)). This selective interference effect therefore raises a theoretical question: when and why can a post-encoding or post-reminder manipulation alter one mode of retrieval more than another, even while both draw on the same recent experience?

One prominent interpretation treats the selective interference effect as evidence for separable memory representations ([Brewin et al., 1996](#); [Brewin, 2014](#)). Under this dual-representation account, traumatic experience produces distinct sensory and contextual memory traces — the first supporting involuntary re-experiencing, the second supporting deliberate recall ([Brewin et al., 2010](#); [Brewin, 2014](#)). A visuospatial interference task competes for the perceptual processing resources needed to consolidate the sensory trace, reducing later intrusions while leaving the contextual trace and the voluntary memory it supports relatively intact ([Holmes et al., 2009](#); [James et al., 2015](#)). When the intervention follows a reminder at longer delays, the account invokes reconsolidation: the reminder reactivates the sensory trace, and the interfering task disrupts its restabilization during a time-limited window ([James et al., 2015](#); [Kindt et al., 2009](#)). The account thus ties selective interference to separate memory representations, modality-specific disruption of one trace, and a discrete reconsolidation window.

Here we argue that selective interference need not be interpreted as evidence for differently vulnerable memory representations. It may instead reflect differences in retrieval processes operating over a shared set of episodic representations. Mechanisms supporting selective, goal-directed retrieval in voluntary recall may be sufficient to sidestep or overcome interference that nonetheless reduces the likelihood of unintentional recall. On this interpretation, the selective-interference effect reflects a capacity to intentionally focus retrieval on goal-relevant event information while reducing the influence of competing memories.

We formalize and evaluate this proposal within a retrieved-context framework drawing on the emotional Context Maintenance and Retrieval model family, including eCMR and CMR3 (Cohen & Kahana, 2022; Talmi et al., 2019). Retrieved-context models explain recall by treating context as a gradually changing state that is updated by studied items, reinstated by recalled items, and used to cue later memory search (Howard & Kahana, 2002; Polyn et al., 2009). Emotional extensions adds source-like emotional context and emotion-dependent binding strength, allowing the same architecture to capture how emotional content changes the organization and success of recall Cohen & Kahana (2022). Variants also model strategic control over memory search, allowing the model to capture how retrieval goals and task demands to shape recall (Cohen & Kahana, 2022; Lohnas et al., 2015; Sederberg et al., 2008). Together, these commitments make the framework a principled setting for testing whether selective interference can arise within a single emotional episodic-memory system.

Consolidation and reconsolidation accounts locate selective interference in a change to the availability of the film representation after encoding or reminder reactivation. A context-binding account shifts the explanatory focus to the retrieval consequences of encoding later experiences in contexts that overlap with the original event (Yonelinas et al., 2019). For selective interference, our model treats the original event, any reminder, and subsequent interference as episodes encoded by binding item representations to a continuously evolving context representation (Howard & Kahana, 2002; Polyn et al., 2009; Yonelinas et al., 2019). Later retrieval is driven by the current context state as a cue, so competition depends on how strongly different episodes are associated with overlapping regions of context. Deliberate recall is more resistant because it combines a more target-oriented initial cue with a sharper decision rule, both of which favor film items over competitors. The dissociation between involuntary and voluntary recall is thus attributed to differences in retrieval state applied to the same stored episodes.

Our simulations evaluate the account at the level where a retrieved-context model makes its distinctive predictions: the distribution and organization of recall across the film, interference, and delay phases of a trauma-film paradigm. They first demonstrate the core selective-interference

dissociation without separately vulnerable representations: an interference task can reduce intrusion-like retrieval more than deliberate recall, even though the same stored film episode supports both. They then distinguish the control mechanisms that protect deliberate recall, showing how a more film-directed retrieval state and a sharper decision rule produce different patterns of impairment in both recall rates and recall organization. The next simulations separate the contributions of competitor strength and contextual overlap, clarifying when an interference task becomes an effective competitor for film retrieval and how reminder-driven reinstatement can support delayed interference. Follow-up simulations use the same framework to interpret heterogeneity across empirical paradigms. They show that film-related cues during intrusion monitoring can narrow the intrusion-voluntary-memory dissociation by making intrusion-like retrieval less purely unguided. They also show that item-specific or film-specific probes can make voluntary access less vulnerable to interference than tests that rely on broad context-to-item search, because broad search leaves film items competing more directly with task competitors. The result is a mechanistic alternative to separately vulnerable memory representations and a set of specific hypotheses about when selective interference should appear and how it should vary across paradigms.

### **A Retrieved-Context Account of Selective Interference**

The account developed here builds on a contextual binding theory of episodic memory ([Howard & Kahana, 2002](#); [Yonelinas et al., 2019](#)). In this framework, each experienced event is encoded by binding it to the current state of a continuously evolving context representation. Context encompasses temporal, spatial, and cognitive features, some of which change rapidly from one event to the next while others shift gradually with ongoing experience ([Yonelinas et al., 2019](#)). Forgetting, on this view, arises primarily from contextual interference rather than time-dependent decay or consolidation. Events encoded in similar context states compete during retrieval, and the more competitors share a target's contextual neighborhood, the less likely the target is to be retrieved ([Mensink & Raaijmakers, 1989](#); [Yonelinas et al., 2019](#)).

Retrieved-context theory formalizes this binding framework with two associative pathways

(Howard & Kahana, 2002; Polyn et al., 2009). Context-to-item associations allow the current context state to cue all events encoded in similar states, producing competitive retrieval in which events vie for recall based on the strength of their contextual match. Item-to-context associations allow a recalled or re-encountered event to reinstate the context in which it was originally experienced, a mechanism absent from earlier context theories. This reinstatement mechanism creates a bidirectional loop: retrieving an event drives context back toward the state present at encoding, which in turn cues neighboring events, producing the temporal contiguity effect (Kahana, 1996; Sederberg et al., 2008).

The critical property for the present account is that context-to-item retrieval is competitive: any event encoded in the current contextual neighborhood is a candidate, and adding new events to that neighborhood increases retrieval competition. In the trauma-film paradigm, intrusive memories arise through this pathway as ongoing context drifts near states associated with the film, cueing film events without intention. When a participant performs a visuospatial task such as Tetris immediately after viewing a distressing film, the events the task produces are encoded in temporal context that overlaps substantially with the film's contextual neighborhood. When context later drifts near the film region, interference events compete alongside film events, reducing the probability that any single film event is retrieved as an intrusion. The strength of this interference depends on how many competitors are encoded, how strongly they are bound, and how much their associated temporal or source context overlaps with the film's. Perceptual modality can matter in this framework, but only insofar as it changes competitor strength or overlap with the film-related context.

The item-to-context pathway extends this account to post-reminder interference. In the trauma-film paradigm, presenting a film clip as a reminder retrieves its associated film context via item-to-context retrieval, pulling the current context back toward the state present during the original viewing (James et al., 2015; Kessler et al., 2020). A visuospatial task performed in this reinstated state encodes competitors that overlap with the film context, enabling interference through the same contextual competition as the immediate case. Interference effectiveness

depends on two continuous variables: how strongly the reminder reinstates film context, and how much subsequent experience has caused the reinstated context to drift before the interference task begins. In the model, delayed interference is therefore governed by graded reinstatement and overlap before competitor encoding.

Item-to-context retrieval is less exposed to contextual competition than context-to-item retrieval. For example, in recognition, the test supplies a film item as the retrieval probe rather than asking the current context to sample among film and task competitors. The probe retrieves its own associated context through item-to-context associations formed during film encoding. Interference items encoded later can add competitors to context-to-item retrieval, but they do not directly overwrite the film item's own item-to-context association. Recognition-like access therefore leaves little room for the contextual-competition mechanism that reduces intrusion-like recall. This protection is different from the graded protection available to deliberate free recall, which still depends on context-to-item retrieval and must use control to favor film items over competitors.

Deliberate recall of the film presents a subtler case. Like intrusions, it uses context-to-item retrieval and is therefore exposed to competitor interference. Two control mechanisms can partially offset this vulnerability. Reinstating the film's starting context as the initial retrieval cue biases recall toward the film region and away from the interference neighborhood; sharpening competition among candidates amplifies strongly associated film events relative to weakly matched competitors. Together, these mechanisms produce graded protection. This protection is weaker than item-anchored access but meaningfully reduces vulnerability, with the degree of sparing depending on how much control the participant exerts.

These mechanisms reproduce several core findings from the episodic memory literature. In free recall, the context-reinstatement loop accounts for temporal contiguity (the tendency to recall temporally neighboring events in sequence) and for the persistence of this pattern across long filled delays ([Howard & Kahana, 2002](#); [Kahana, 1996](#); [Sederberg et al., 2008](#)). Memories from recent but irrelevant experiences also intrude during recall of a target episode, with a

frequency that decays gradually over intervening time (Howard et al., 2008; Lohnas et al., 2015; Zaromb et al., 2006). Extensions of the model (eCMR, CMR3) add emotional arousal and valence as dimensions of context, so that high-arousal events are bound to temporal context and to an emotional context reflecting the affective state at encoding (Talmi et al., 2019). This extension is directly relevant to the present work, as traumatic film material is encoded with strong arousal, and the model's treatment of arousal as context captures the enhanced encoding and distinctive retrieval dynamics that characterize emotional material.

The account therefore replaces a storage-level dissociation with a retrieval-level one. Intrusion-like and deliberate recall operate over the same item-context associations, but they begin from different retrieval states and differ in how strongly retrieval is constrained toward the film. Interference strength is determined by the number and strength of task competitors and by their overlap with film-related temporal, source, and emotional context. Post-reminder effects arise when a reminder reinstates film context before competitor encoding. These commitments define the model components specified next: phase-specific encoding, reminder-driven reinstatement, and retrieval-mode settings.

## **Model and Paradigm**

### **Simulation Overview**

#### **Results**

#### **Simulation 1: Core Selective-Interference Dissociation**

Premise: The basic selective-interference result is a 2 x 2 contrast between interference condition and retrieval configuration. The section asks whether the same stored film episode and the same post-film interference manipulation can produce larger impairment in intrusion-like retrieval than in deliberate recall.

Intended text take-away: This section establishes the core sufficiency result. The model can reproduce the qualitative intrusion-voluntary-memory dissociation without separately weakening film items for one measure. Deliberate recall may still be impaired; the claim is relative protection, not immunity.

**Simulation 2: Control Mechanisms Underlying Deliberate-Recall Protection**

Premise: Deliberate recall is not categorically protected. Its relative sparing depends on control operations that make retrieval more film-directed.

Intended text take-away: This section earns the “relatively intact voluntary memory” claim. Sparing is a graded consequence of retrieval control rather than a property attached to voluntary tests as a category. Weakening these control operations should make deliberate recall resemble intrusion-like retrieval more closely.

**Simulation 3: Competitor Strength, Contextual Overlap, and Emotional-Context****Amplification**

Premise: An interference task matters when it encodes competitors that are both strong enough and contextually close enough to film items. This section asks what makes a task Tetris-like in model terms.

Intended text take-away: This section explains what makes interference effective. The account should not reduce to generic distraction, list length, or task difficulty. Competitor strength matters because competitors are retrievable from film-related context, and shared emotional or source context is relevant only insofar as it increases that competition.

**Simulation 4: Reminder Reinstatement and Delayed Interference**

Premise: The delayed-reminder case applies the overlap mechanism after context has drifted away from the film. A reminder matters because it reinstates film context before competitor encoding.

Intended text take-away: This section connects the model to post-reminder interventions. The reminder is not treated as a special storage-disruption device. It restores the contextual condition required for later competitor encoding to affect film-item retrieval.

**Simulation 5: Heterogeneity Across Recall-Like Test Formats**

Premise: Intrusion diaries, vigilance-style monitoring, free recall, and cued recall differ in how much they rely on broad context-to-item search versus film-directed retrieval support. This

section maps the model's retrieval-control mechanisms onto variation across recall-like empirical measures.

Intended text take-away: This section interprets heterogeneity across empirical paradigms without treating intrusion and voluntary memory as fixed computational kinds. The model predicts a continuum of vulnerability across recall-like tests depending on how much the task supports film-directed retrieval.

### **Simulation 6: Probe-Driven Access as a Boundary Case**

Premise: Item-specific probes are a limiting case of voluntary access because they begin closer to the target item than broad recall does. This section preserves the useful part of the old recognition result without claiming categorical recognition immunity.

Intended text take-away: This section handles film-specific probes and recognition-like measures as a boundary case. The claim is that probe-driven access reduces exposure to competitor competition, not that recognition is absolutely immune or that all voluntary memory has the same protection.

## **General Discussion**

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